Technical Report: Computational Analysis of Username Classes

Dataset characterization, character-level modeling, and cross-validated classification

Prepared for computational linguistics Capstone (See Below for Fall 2026 extension)

Executive summary

This report documents the computational work completed to date on a three-class username dataset. Class A contains documented criminal-associated usernames, Class B contains public-figure usernames, and Class C is a large general Reddit username corpus. The present modeling task is binary rather than three-way: Class A is compared separately with Class B and Class C using character-level TF-IDF and logistic regression. Five-fold stratified cross-validation indicates stronger separation for Class A versus Class B (mean F1 = 0.655; mean PR-AUC = 0.769) than for Class A versus Class C (mean F1 = 0.494; mean PR-AUC = 0.613). These models quantify similarity between observed username classes; they do not infer criminal status. A subsequent supplementary analysis examines additional non-gender orthographic tests and recorded gender metadata within Class B. The Class B gender supplement contains 18,295 platform-identity-deduplicated observations bearing exact male/female labels; seven of 20 platform-specific comparisons survived Holm correction. Those associations were small and are not person-level demographic predictions. Supplemental A/B diagnostics are described as implemented but not independently rerun. A completed three-class NER supplement also tests raw and lightly normalized usernames; normalized detection is highest in Class B (74.65%), but detections have no human-verified accuracy benchmark.

1. Computational objective

The objective is to determine whether orthographic information contained in usernames provides measurable discriminatory signal between Class A and non-Class-A comparison populations. The baseline system intentionally uses a simple, interpretable text-classification pipeline so that later experiments can measure the incremental value of structural username features, alternative comparison samples, and model tuning.

2. Dataset architecture

2.1 Class A

The Class A source contains 236 records and 234 unique usernames, with two duplicate usernames. The classifier uses the cleaned unique username set. Available metadata include age, sex, nationality, crime type, source information, and review fields, although the current classifier uses only the username.

Statistic	Class A
Total records	236
Unique usernames	234
Duplicate usernames	2

Mean username length	8.483
SD username length	3.703
Median length	8
Length range	1-22
Mean digit count	0.517
Mean letter count	7.739
Mean vowel count	2.679
Starts with letter	98.29%
Contains underscore	3.42%
Contains other special character	13.68%
All lowercase	19.66%
All uppercase	8.55%
Contains digit	19.66% (46/234)
Ends with digit	14.53% (34/234)
Repeated character 3+	0.85% (2/234)
Mixed case	70.94% (166/234)

Class A descriptive measures were recalculated directly from the uploaded master CSV after removing missing usernames and exact duplicate username strings (n = 234). The figures in Table 1 reflect this single reproducible calculation basis.

2.2 Class B

Class B represents public-figure usernames. The platform-level descriptive output currently available covers 10,000 X/Twitter observations and 10,000 YouTube-handle observations. The A-vs-B classifier uses a cleaned Class B pool containing 19,750 unique usernames and draws a reproducible random sample of 234 to match the 234 Class A usernames.

Feature	X/Twitter	YouTube
N	10,000	10,000
Unique handles	9,970	9,987
Mean length	10.803	12.219
SD length	2.822	4.166
Median length	11	12
Mean letter count	10.366	11.621
Repeated character 3+	1.17%	1.15%
All lowercase	67.46%	76.40%
All uppercase	1.17%	1.00%

Mixed case	31.15%	19.99%
CamelCase	21.81%	15.77%
Numeric only	0.22%	0.08%
Contains given name	64.95%	74.38%
Contains family name	68.42%	72.48%
Contains full name	41.55%	60.88%

2.3 Class C

Class C is drawn directly from the Kaggle repository colinmorris/reddit-usernames. The source file users.csv stores the Reddit username in the author column. Full-corpus descriptive statistics were computed in chunks of 500,000 rows to avoid loading the entire corpus into memory at once. The resulting analysis covers 25,865,739 non-missing usernames.

Feature	Class C
N usernames	25,865,739
Mean length	11.017
SD length	3.684
Contains digit	29.465%
Ends with digit	24.902%
Starts with letter	98.048%
Contains underscore	10.847%
Contains other special	2.389%
All lowercase	52.230%
All uppercase	2.094%
Mixed case	44.812%
CamelCase pattern	44.373%
Repeated character 3+	2.670%
Mean digit count	0.789
Mean letter count	10.039
Mean vowel count	3.622
Mean delimiter-piece count	1.182

3. Descriptive comparison of username form

The descriptive statistics show substantial differences in username form before classification is attempted. Class A usernames are shorter on average (8.48 characters) than Class C usernames (11.02) and both reported Class B platform samples (10.80 for X/Twitter; 12.22 for YouTube). Class A is also far less frequently all lowercase (19.66%) than Class C (52.23%), X/Twitter Class B (67.46%), or YouTube Class B (76.40%). Conversely, the current Class A output shows a higher rate of other special characters (13.68%) than Class C (2.39%). The newly verified Class A frequencies are 19.66% containing any digit, 14.53% ending in a digit, 0.85% containing three consecutive identical characters, and 70.94% combining uppercase and lowercase letters. They are calculated on the 234 unique nonmissing Class A usernames, and should not be interpreted as evidence of individual criminal involvement.

These descriptive contrasts are not themselves evidence of class predictability. They motivate the classification analysis and provide candidate structural features for later modeling. Comparisons are also subject to source differences: Class B is platform-specific, Class C is Reddit-derived, and Class A is assembled from documented cases. Platform and sampling effects may therefore contribute to observed orthographic differences.

4. Computational preprocessing

The preprocessing pipeline is implemented in Python with pandas. For each class, the relevant username field is selected, missing values are removed, values are converted to strings, and within-class duplicates are removed. Binary labels are assigned with Class A = 1 and the comparison class = 0. Comparison samples are drawn reproducibly using random_state=42, concatenated with Class A, and shuffled.

Class C ingestion is automated with kagglehub. Because the source contains more than 25 million rows, full-corpus descriptive statistics are computed by streaming the file in chunks. The classification experiment does not train on the full corpus: it uses a small random Class C comparison sample.

5. Feature representation

The baseline representation is character-level term frequency-inverse document frequency (TF-IDF). TfidfVectorizer is configured with analyzer='char', ngram_range=(2, 5), and lowercase=True. Thus, each username is represented by weighted character sequences of length two through five. This representation captures recurring sub-string patterns, digit-letter combinations, orthographic fragments, and short character sequences without requiring a predefined linguistic lexicon.

6. Classification model

The baseline classifier is logistic regression with max_iter=1000 and random_state=42. Balanced class weighting is used for the A-vs-C experiment because the comparison sample contains 300 Class C usernames versus 234 Class A usernames. The A-vs-B experiment is exactly balanced at 234 usernames per class and uses standard weighting.

7. Evaluation procedure

Initial experiments used stratified 80/20 train/test splits. The primary evaluation was then changed to five-fold stratified cross-validation because only 234 Class A usernames are available. In every fold, the TF-IDF vectorizer is fitted exclusively on the training partition, the held-out usernames are transformed using that fitted vectorizer, and a new logistic-regression model is trained. This avoids feature-space leakage from the held-out fold.

Metrics include accuracy, Class A precision, Class A recall, Class A F1-score, and PR-AUC/average precision. Precision measures the proportion of Class A predictions that are correct; recall measures the proportion of held-out Class A usernames detected; F1 summarizes the precision-recall balance; PR-AUC evaluates ranking quality for the positive Class A label.

8. Class A versus Class B

The A-vs-B modeling dataset contains 468 usernames: 234 Class A and 234 randomly sampled Class B usernames. Five-fold stratified cross-validation produced the following results.

Fold	Accuracy	A Precision	A Recall	A F1	PR-AUC
1	0.798	0.804	0.787	0.796	0.842
2	0.617	0.617	0.617	0.617	0.691
3	0.617	0.657	0.489	0.561	0.749

4	0.710	0.794	0.574	0.667	0.773
5	0.677	0.722	0.565	0.633	0.789

Metric	Mean	SD
Accuracy	0.684	0.067
Class A precision	0.719	0.074
Class A recall	0.607	0.099
Class A F1	0.655	0.078
PR-AUC	0.769	0.050

The model shows moderate discrimination between Class A and Class B. Mean Class A precision is 0.719, mean recall is 0.607, and mean F1 is 0.655. Mean PR-AUC is 0.769. Performance varies across folds, with Class A recall ranging from 0.489 to 0.787 and F1 from 0.561 to 0.796, indicating sensitivity to the specific Class A examples held out in each fold.

9. Class A versus Class C

The reported A-vs-C modeling dataset contains 534 usernames: all 234 Class A usernames and a reproducible random sample of 300 Class C Reddit usernames. Five-fold stratified cross-validation produced the following results.

Fold	Accuracy	A Precision	A Recall	A F1	PR-AUC
1	0.598	0.559	0.404	0.469	0.595
2	0.626	0.600	0.447	0.512	0.623
3	0.664	0.649	0.511	0.571	0.736
4	0.579	0.521	0.532	0.526	0.546
5	0.585	0.538	0.304	0.389	0.566

Metric	Mean	SD
Accuracy	0.610	0.031
Class A precision	0.573	0.046
Class A recall	0.440	0.081
Class A F1	0.494	0.062
PR-AUC	0.613	0.067

The A-vs-C classifier exhibits weaker discrimination. Mean Class A recall is 0.440, so fewer than half of held-out Class A usernames are detected at the default decision threshold. Mean F1 is 0.494 and mean PR-AUC is 0.613. The model therefore captures some character-level signal but substantial overlap remains between Class A and the sampled Reddit comparison usernames.

10. Direct comparison of classification experiments

Metric	A vs. B	A vs. C	Difference (B - C)
Accuracy	0.684	0.610	+0.074
Class A precision	0.719	0.573	+0.146
Class A recall	0.607	0.440	+0.167
Class A F1	0.655	0.494	+0.161
PR-AUC	0.769	0.613	+0.156

Using the same basic character-level representation and logistic-regression architecture, Class A is more readily separated from Class B than from Class C on every reported mean metric. The largest absolute difference is Class A recall (+0.167), followed by F1 (+0.161) and PR-AUC (+0.156).

A plausible computational hypothesis is that the general Reddit corpus contains greater orthographic diversity and therefore overlaps more extensively with Class A than the public-figure comparison set. However, the current experiments do not isolate platform, demographic, or sampling effects, so the cause of the performance difference should not be inferred from classifier performance alone.

11. Full-corpus Class C feature computation

The Class C descriptive pipeline is designed for scale. Candidate CSV/TSV/TXT files are detected in the downloaded Kaggle directory, the largest source file is selected, and the username column is detected from common names including username, user, author, and name. The file is processed in 500,000-row chunks.

A first pass accumulates a username-length histogram and running sums to calculate mean, sample standard deviation, and empirical quartile cutoffs without storing all usernames in memory. A second pass computes binary and count features for each chunk and aggregates them overall, by length quartile, and by one-standard-deviation length bands. Output tables are saved as CSV files for reproducibility.

12. Supplementary non-gender statistical analysis

Purpose. An expanded Class A versus Class B notebook adds explicit structural comparisons to the primary character-level classifier. The new analyses consider username length, digit frequency, terminal digits, underscores, punctuation, capitalization, repeated characters, and related orthographic properties. These analyses extend, rather than replace, the original three-corpus comparison and A-vs-B/A-vs-C classification results.

Methods. Continuous or count characteristics are compared using Mann–Whitney U tests, while binary variables use Fisher's exact or chi-square procedures depending on cell counts. Rank-biserial correlations or categorical effect-size measures accompany significance tests; Holm and Benjamini–Hochberg adjustments address multiplicity. Further classifier diagnostics include out-of-fold performance, majority-class comparisons, ROC-AUC, precision-related metrics, calibration losses, Matthews correlation, Wilson intervals, and McNemar comparisons, subject to the outputs available from the notebook.

Verification status. Class A descriptive statistics have been recalculated from the source CSV using 234 unique usernames. The supplementary A/B inferential notebook has not yet been rerun end-to-end against the complete source files; accordingly, tests without verified outputs are described as methods rather than numerical findings.

13. Supplementary gender-associated analysis within Class B

Dataset and inclusion. The Class B source contains 20,000 X and YouTube records. Of these, 19,089 have a nonmissing recorded gender string. The inferential analysis deduplicates by platform and Wikidata identifier, then uses 18,295 records with exact "male" or "female" labels. Other gender strings and missing entries are retained in the data-quality accounting but not relabeled into either binary group. Gender is a documented dataset field, not inferred from username spelling.

Statistical design. Ten orthographic characteristics are tested separately on each platform (20 comparisons): length, digit count, punctuation count, any digit, ending digit, underscore, uppercase letter,

mixed case, year-like number, and repeated-character run of three or more. Mann–Whitney U is applied to numeric/count measures; chi-square to binary features. Holm-corrected p-values are the decision criterion, with alpha = 0.05; effect sizes help separate detectable from substantively meaningful differences.

Platform	Feature	Male	Female	Holm-adjusted p
X	Mean username length	10.560	11.133	2.13×10^{-16}
YouTube	Mean username length	12.005	12.682	8.01×10^{-14}
X	Underscore presence	14.57%	17.49%	0.00261
YouTube	Underscore presence	4.77%	6.82%	0.000554
X	Long repetition (3+)	0.86%	1.92%	0.000180

Selected significant findings (all comparisons above are platform-specific). Male/female labels denote the source metadata, not classifications made from usernames. Source: Class_B_Gender_Supplement_Statistics.csv.

Across all 20 gender-associated comparisons, seven reached Holm-adjusted $p < 0.05$. In addition to the five shown in the table, punctuation count differed significantly on X and YouTube (Holm-adjusted $p \approx 0.00216$ on each). Female-labeled accounts had slightly longer usernames on both platforms; rank-biserial correlations were approximately -0.106 (X) and -0.098 (YouTube), indicating small effects. The results should not be converted into individual-level gender estimates.

14. Supplementary three-class named-entity recognition (NER)

A completed secondary experiment examined whether spaCy 3.8.14 with en_core_web_trf assigned at least one selected named-entity label to usernames. The analysis included 234 distinct Class A usernames, 19,957 distinct Class B platform–username records, and a seeded sample of 19,957 Class C usernames drawn from the 25,865,739-record Reddit corpus. Input was analyzed both raw and after light normalization (leading @ removal, CamelCase and delimiter splitting, letter–digit boundary splitting, and punctuation cleanup). The analytic endpoint was detection, not independently validated NER correctness.

14.1 Detection rates and platform comparison

Population	N	Raw detection	Normalized detection
Class A	234	133 (56.84%)	128 (54.70%)
Class B	19,957	13,754 (68.92%)	14,897 (74.65%)
Class C	19,957	11,051 (55.37%)	10,808 (54.16%)
Class B: X	9,970	7,200 (72.22%)	8,051 (80.75%)
Class B: YouTube	9,987	6,554 (65.63%)	6,846 (68.55%)

Class B had the highest detection rates, particularly after normalization. The Class B difference between X and YouTube is descriptive; the package does not demonstrate its cause. PERSON was the most common model-assigned entity label, but a PERSON prediction is not evidence of verified identity or a correct personal name.

14.2 Inferential analysis

For normalized inputs, Holm-adjusted Fisher exact tests found differences for A versus B ($p = 2.67 \times 10^{-10}$; Cohen's $h = -0.421$) and B versus C (reported p underflowed to 0; $h = 0.432$), but not A versus C (adjusted

p = 1.000; h = 0.011). Raw-input comparisons showed the same broad pattern: A versus B adjusted p = 0.000355, B versus C adjusted p ≈ 4.18×10^{-171}, and A versus C adjusted p = 1.000. These tests describe differences in NER detection rates, not differences in true named entities.

Exact paired McNemar tests compared raw and normalized versions of the same usernames. Normalization changed detection significantly for B (1,894 gained versus 751 lost; Holm-adjusted p ≈ 2.30×10^{-112}) and C (1,817 gained versus 2,060 lost; adjusted p ≈ 0.000202), but not A (12 gained versus 17 lost; adjusted p ≈ 0.458). A statistically significant change need not indicate improved recognition accuracy.

14.3 Reproducibility and validation boundary

The supplied NER_Complete_Statistics_Package contains a detailed HTML report, configuration, input audit, class/platform and label summaries, Fisher and McNemar tables, graphs, and reproduce_ner_statistics.py. The reference and reproduced aggregate summaries match. The script reconstructs aggregate rates, Wilson confidence intervals, tests, and visualizations from saved counts. It does not rerun the transformer on the original username samples. The exact Class C sample and original inference environment were not retained, and no human-labeled ground truth was available to calculate NER precision, recall, or F1. Thus these are reproducible aggregate results but not an independently reproduced model-inference run.

NER source: NER_Complete_Statistics_Package(1).zip (NER_Report.html, ner_configuration.json, reference_outputs/*, reproduced_outputs/*, reproduce_ner_statistics.py).

15. Integrated interpretation and verification boundaries

The original research has three distinct empirical components: (i) descriptive study of Class A, Class B and the complete 25,865,739-name Class C Reddit corpus; (ii) binary classification of A versus sampled B and of A versus sampled C; and (iii) additional structural and gender-associated analyses. Class C's full-corpus statistics are descriptive; the A-vs-C classifier used 300 sampled Class C usernames, not all 25.9 million.

Differences between public-figure accounts, documented criminal-associated aliases, and Reddit names may be influenced by platform constraints, visibility, occupations, age distributions, sampling, and documentation practices. Gender results describe associations among recorded categories in the public-figure Class B sample and must not be generalized to all users or used to assign gender to unlabelled aliases.

Class A descriptive statistics in this report were recalculated from the uploaded master CSV (234 unique, nonmissing usernames). The original classifier evaluation results are retained as previously reported; the classification pipeline was not rerun as part of this descriptive update.

16. Limitations

The primary modeling limitation is the small Class A sample. Only 234 unique Class A usernames are available, so the model has limited examples from which to learn the range of Class A variation. Increasing the number of comparison usernames alone cannot solve this positive-class limitation.

Second, the datasets are not platform-matched. Class C is Reddit-derived, Class B is represented here by X/Twitter and YouTube, and Class A is assembled from documented cases that may originate from

multiple platforms or contexts. Some classification signal may therefore reflect platform conventions rather than the conceptual class distinction.

Third, the original model uses only character n-grams. Explicit structural variables are included in the descriptive analysis but are not yet incorporated into the original classifier. The supplementary inferential A/B tests remain subject to end-to-end reproduction.

17. Planned computational extensions

The next modeling stage will compare three feature configurations: character TF-IDF alone, explicit structural username features alone, and a combined TF-IDF plus structural model. Candidate structural features include username length, digit count and proportion, terminal digits, underscores, other special characters, capitalization pattern, CamelCase, repeated characters, letter count, vowel count, and delimiter structure.

Additional experiments can test alternative character n-gram ranges, tune logistic-regression regularization, compare a linear support-vector classifier, examine decision thresholds when recall is prioritized, and test the stability of A-vs-C performance across multiple random Class C samples and sample sizes. Expansion of Class A would provide the most direct improvement in the amount of positive-class training information.

18. Current computational conclusion

The current pipeline demonstrates that username strings contain measurable class-related signal under a simple character-level model, but the strength of that signal depends strongly on the comparison population. Class A versus Class B produces moderate cross-validated discrimination, whereas Class A versus Class C is substantially weaker. The results support continued computational investigation of username structure while also showing that character n-grams alone are insufficient for robust individual-level classification. A later Class B gender supplement identified small population-level associations in username length and selected structural features. These results do not identify a particular user's gender and do not change the conclusions from the original three-corpus experiments. Complementary NER analysis finds that entity-label detection is particularly frequent in Class B, although predicted labels cannot be treated as verified identities or recognition accuracy.

AI assistance disclosure: ChatGPT was used for code troubleshooting, supplementary-analysis workflow assistance, document formatting, and report integration. The Class A/B and A/C metrics reported above are taken from the original technical report; supplementary gender test results are drawn from the separately saved statistical output. The available materials do not permit independent rerunning of every computational experiment.